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MIS581 – Capstone Research Project Submission

YouTube Video Link

<https://youtu.be/kWyRi5bMlNo>

GitHub Repository Link

[Sparky-Clark/MIS581: Capstone Research Project](#)

Screenshots of Data Analysis Outcomes

Figure 1

First 5 Rows of Imported Dataset

```
In [2]: # Display first 5 rows to view basic structure
df.head()
```

Out[2]:

	age	gender_female	dispo_post_acute	dispo_transfer	dispo_ama	los_days_capped	infection_flag	adm_dx_sepsis	adm_dx_pna	adm_dx_hf	...	med_risk_high	ed
0	30	0	0	0	0	2	0	0	0	0	...	0	
1	73	0	0	0	0	3	0	0	0	0	...	0	
2	81	1	0	0	0	2	0	0	0	1	...	0	
3	83	0	1	0	0	5	0	0	0	0	...	0	
4	75	0	0	0	0	3	0	0	0	0	...	0	

5 rows × 27 columns

Figure 2

Dataset and Variable Information

```
In [3]: # Check data types and null value counts for each variable
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 26014 entries, 0 to 26013
Data columns (total 27 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   age                                   26014 non-null  int64
1   gender_female                         26014 non-null  int64
2   dispo_post_acute                     26014 non-null  int64
3   dispo_transfer                       26014 non-null  int64
4   dispo_ama                           26014 non-null  int64
5   los_days_capped                     26014 non-null  int64
6   infection_flag                       26014 non-null  int64
7   adm_dx_sepsis                       26014 non-null  int64
8   adm_dx_pna                          26014 non-null  int64
9   adm_dx_hf                           26014 non-null  int64
10  adm_dx_copd                         26014 non-null  int64
11  adm_dx_rf                           26014 non-null  int64
12  adm_dx_aki                          26014 non-null  int64
13  mort_risk_medium                    26014 non-null  int64
14  mort_risk_high                      26014 non-null  int64
15  mort_risk_missing                   26014 non-null  int64
16  med_risk_medium                     26014 non-null  int64
17  med_risk_high                       26014 non-null  int64
18  ed_low                             26014 non-null  int64
19  ed_moderate                         26014 non-null  int64
20  ed_high                             26014 non-null  int64
21  ip_admit_utilization_low            26014 non-null  int64
22  ip_admit_utilization_moderate       26014 non-null  int64
23  ip_admit_utilization_high           26014 non-null  int64
24  icu_admit_flag                      26014 non-null  int64
25  rrt_flag                            26014 non-null  int64
26  readmit_30d                        26014 non-null  int64
dtypes: int64(27)
memory usage: 5.4 MB
```

Figure 3

Descriptive Statistics of Discrete Numerical Variables

```
In [4]: # Display descriptive stats of the discrete numeric variables  
df[['age', 'los_days_capped']].describe().round(2)
```

Out[4]:

	age	los_days_capped
count	26014.00	26014.00
mean	55.25	3.78
std	20.97	4.08
min	18.00	0.00
25%	35.00	2.00
50%	59.00	3.00
75%	73.00	4.00
max	104.00	30.00

Figure 4

Side-by-Side Histograms for Age and LOS Variables

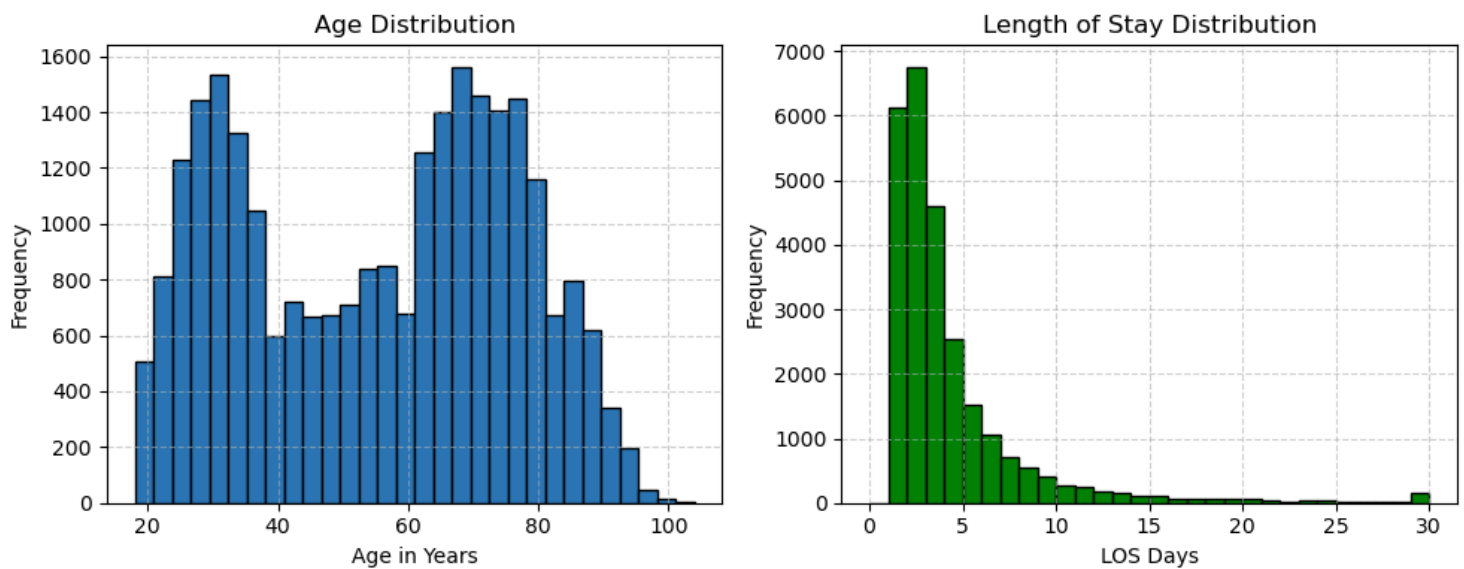


Figure 5*Bivariate Analyses of Binary Indicator Variables*

	Variable Group	Category	Count	% of Total	Readmission Rate
0	Gender	gender_female	15364	59.1	8.2
1	Gender	(Reference) gender_male	10650	40.9	10.8
2	Clinical Indicators	infection_flag	1720	6.6	19.2
3	Clinical Indicators	(Reference) no_infection	24294	93.4	8.6
4	Clinical Indicators	icu_admit_flag	3815	14.7	12.7
5	Clinical Indicators	(Reference) no_icu_admit	22199	85.3	8.7
6	Clinical Indicators	rrt_flag	2352	9.0	13.1
7	Clinical Indicators	(Reference) no_rrt	23662	91.0	8.9

Figure 6*Bivariate Analyses of Admission Diagnosis Binary Indicator Variables*

	Variable Group	Category	Count	% of Total	Readmission Rate
0	Admission Diagnoses	adm_dx_sepsis	611	2.3	14.2
1	Admission Diagnoses	(Reference) no_sepsis	25403	97.7	9.2
2	Admission Diagnoses	adm_dx_pna	1114	4.3	11.3
3	Admission Diagnoses	(Reference) no_pna	24900	95.7	9.2
4	Admission Diagnoses	adm_dx_hf	739	2.8	14.9
5	Admission Diagnoses	(Reference) no_hf	25275	97.2	9.1
6	Admission Diagnoses	adm_dx_copd	472	1.8	14.6
7	Admission Diagnoses	(Reference) no_copd	25542	98.2	9.2
8	Admission Diagnoses	adm_dx_rf	1370	5.3	12.3
9	Admission Diagnoses	(Reference) no_rf	24644	94.7	9.1
10	Admission Diagnoses	adm_dx_aki	976	3.8	14.1
11	Admission Diagnoses	(Reference) no_aki	25038	96.2	9.1

Figure 7*Bivariate Analyses of One-Hot Encoded Binary Variables*

	Variable Group	Category	Count	% of Total	Readmission Rate
0	Discharge Disposition	(Reference) dispo_home	21984	84.5	8.2
1	Discharge Disposition	dispo_post_acute	3382	13.0	12.2
2	Discharge Disposition	dispo_transfer	217	0.8	49.8
3	Discharge Disposition	dispo_ama	431	1.7	18.6
4	Mortality Risk	(Reference) mort_risk_low	15492	59.6	6.7
5	Mortality Risk	mort_risk_medium	3717	14.3	14.3
6	Mortality Risk	mort_risk_high	2566	9.9	18.7
7	Mortality Risk	mort_risk_missing	4239	16.3	8.8
8	Medication Risk	(Reference) med_risk_low	13937	53.6	6.0
9	Medication Risk	med_risk_medium	9790	37.6	9.6
10	Medication Risk	med_risk_high	2287	8.8	28.0
11	ED Utilization	(Reference) ed_none	14784	56.8	6.7
12	ED Utilization	ed_low	7654	29.4	8.8
13	ED Utilization	ed_moderate	2615	10.1	18.0
14	ED Utilization	ed_high	961	3.7	28.6
15	Inpatient Utilization	(Reference) ip_admit_utilization_none	6809	26.2	6.6
16	Inpatient Utilization	ip_admit_utilization_low	15068	57.9	5.5
17	Inpatient Utilization	ip_admit_utilization_moderate	3332	12.8	24.8
18	Inpatient Utilization	ip_admit_utilization_high	805	3.1	38.5

Correlation Matrix of All Predictor Variables

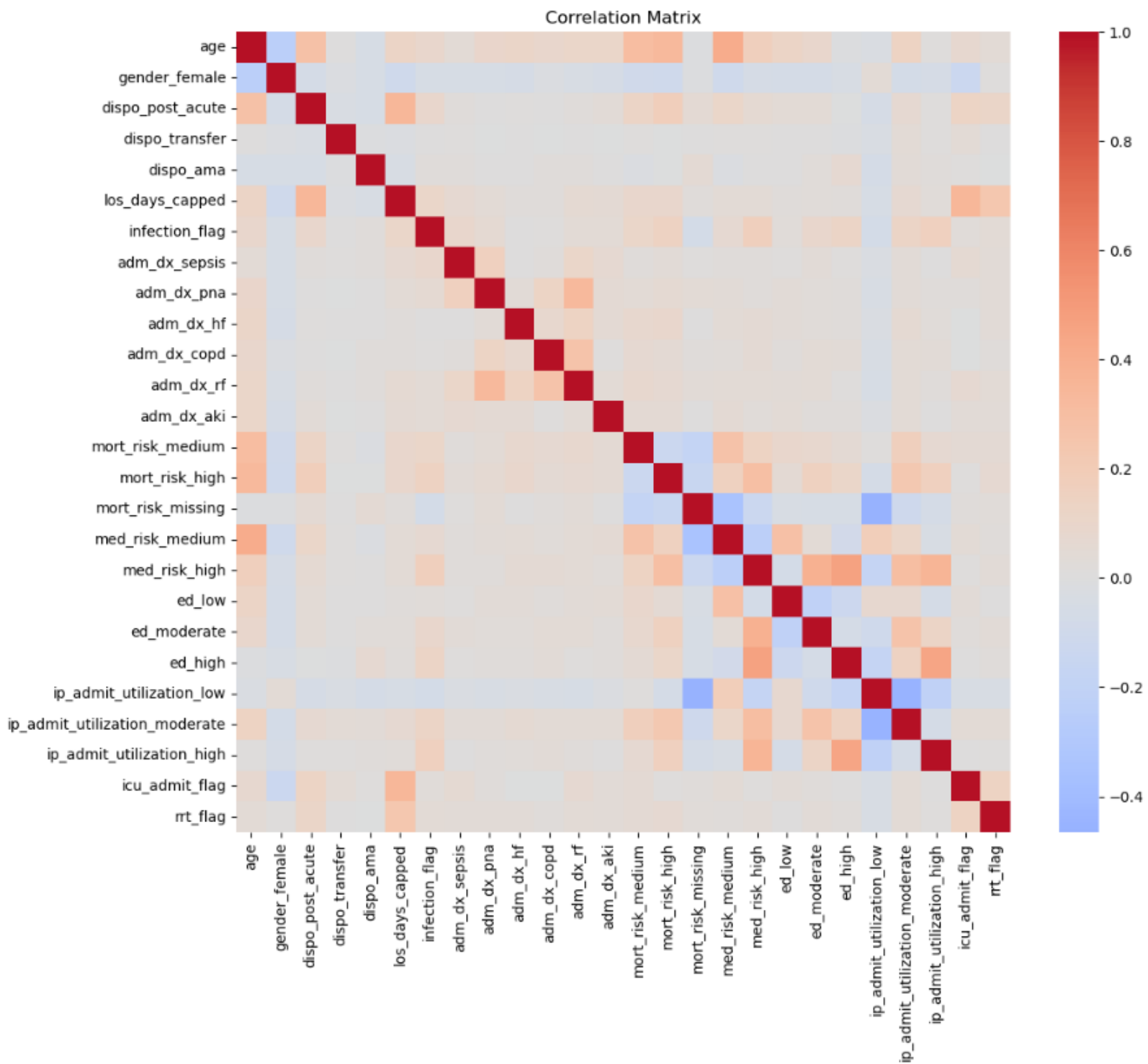


Figure 8

Variance Inflation Factor (VIF) Table for All Predictor Variables

	Feature	VIF
18	med_risk_high	2.565690
22	ip_admit_utilization_low	2.282701
17	med_risk_medium	2.114802
23	ip_admit_utilization_moderate	2.052613
16	mort_risk_missing	2.012789
1	age	1.808526
21	ed_high	1.794962
24	ip_admit_utilization_high	1.644281
20	ed_moderate	1.614120
15	mort_risk_high	1.500663
14	mort_risk_medium	1.403825
6	los_days_capped	1.353173
19	ed_low	1.280529
3	dispo_post_acute	1.248196
12	adm_dx_rf	1.220844
25	icu_admit_flag	1.172682
9	adm_dx_pna	1.158719
2	gender_female	1.103932
7	infection_flag	1.088442
11	adm_dx_copd	1.085613
26	rrt_flag	1.079728
8	adm_dx_sepsis	1.048236
10	adm_dx_hf	1.046265
5	dispo_ama	1.025497
13	adm_dx_aki	1.023507
4	dispo_transfer	1.009682

Figure 9

Logistic Regression AUC and Confusion Matrix Values with 0.5 Threshold

Out[12]: LogisticRegression(class_weight='balanced', max_iter=1000)

**In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.**

```
In [13]: ## Make predictions
y_pred = lr.predict(X_test)
y_prob = lr.predict_proba(X_test)[:, 1]

## Import AUC function
from sklearn.metrics import roc_auc_score

## Calculate AUC value
auc = roc_auc_score(y_test, y_prob)

## Display AUC value
print("LR AUC:", auc)
```

LR AUC: 0.7467548946250216

```
In [14]: ## Import confusion matrix function
from sklearn.metrics import confusion_matrix

## Create confusion matrix
cm_lr = confusion_matrix(y_test, y_pred)

## Display results
print("LR Confusion Matrix (Threshold = 0.5)")
print("_____")
print(cm_lr)
```

LR Confusion Matrix (Threshold = 0.5)

```
[[5759 1322]
 [ 318  406]]
```


Figure 10

Logistic Regression Classification Report with 0.5 Threshold

```
In [15]: ## Import the classification report function
from sklearn.metrics import classification_report

## Create and display a classification report
print("LR Classification Report (Threshold = 0.5)")
print("_____")
print(classification_report(y_test, y_pred))
```

LR Classification Report (Threshold = 0.5)

	precision	recall	f1-score	support
0	0.95	0.81	0.88	7081
1	0.23	0.56	0.33	724
accuracy			0.79	7805
macro avg	0.59	0.69	0.60	7805
weighted avg	0.88	0.79	0.82	7805

Figure 11

Logistic Regression ROC Curve

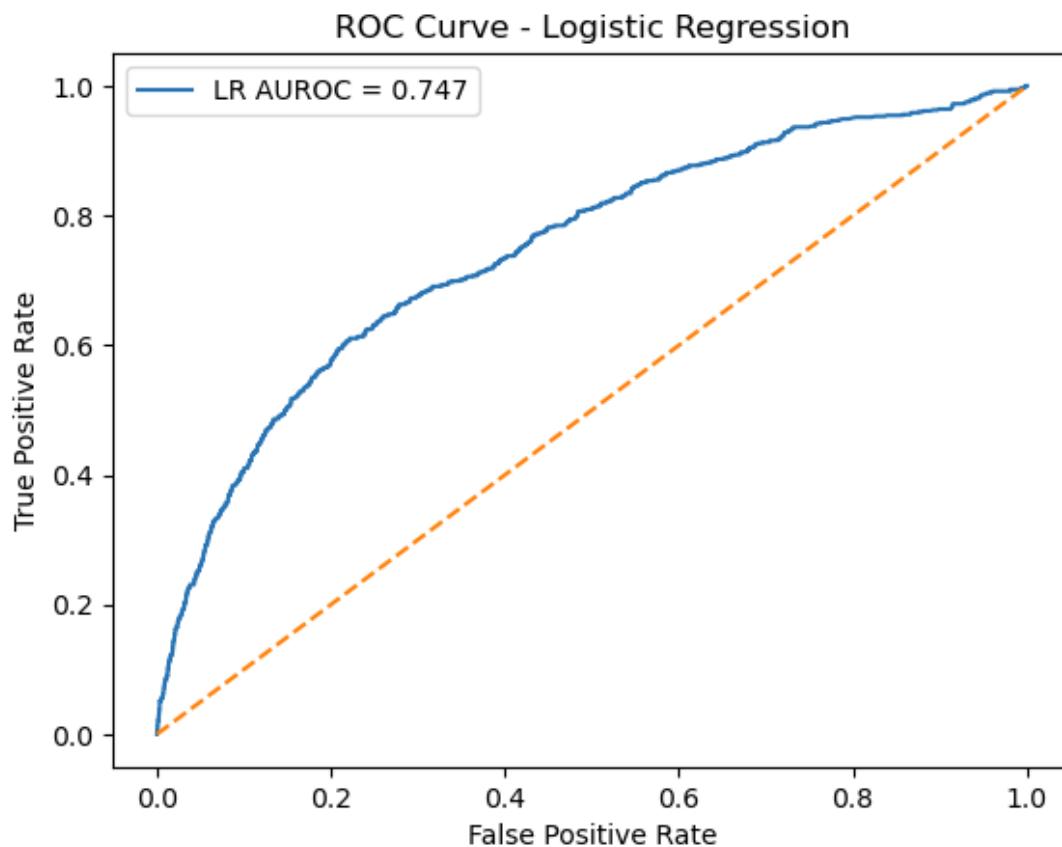


Figure 13*Logistic Regression Feature Importance Table*

	LR Feature	Coefficient	Odds Ratio
3	dispo_transfer	2.506240	12.258750
23	ip_admit_utilization_high	1.860507	6.426996
22	ip_admit_utilization_moderate	1.470488	4.351357
17	med_risk_high	1.330611	3.783354
4	dispo_ama	0.797388	2.219735
15	mort_risk_missing	0.716002	2.046235
16	med_risk_medium	0.462759	1.588451
7	adm_dx_sepsis	0.276104	1.317984
12	adm_dx_aki	0.260084	1.297039
9	adm_dx_hf	0.238570	1.269433
10	adm_dx_copd	0.163901	1.178097
13	mort_risk_medium	0.146930	1.158273
6	infection_flag	0.107301	1.113269
14	mort_risk_high	0.095737	1.100470
21	ip_admit_utilization_low	0.076827	1.079855
25	rrt_flag	0.063510	1.065570
5	los_days_capped	0.054230	1.055727
0	age	0.000937	1.000937
1	gender_female	-0.013881	0.986215
24	icu_admit_flag	-0.014041	0.986057
11	adm_dx_rf	-0.032036	0.968472
2	dispo_post_acute	-0.095183	0.909207
8	adm_dx_pna	-0.096689	0.907838
18	ed_low	-0.188095	0.828536
19	ed_moderate	-0.240183	0.786484
20	ed_high	-0.347063	0.706761

Figure 13

Logistic Regression Feature Importance Table

```
In [18]: ## Check 0.3 classification threshold
y_pred_03 = (y_prob >= 0.3).astype(int)

## Create a confusion matrix with 0.3 threshold
cm_03 = confusion_matrix(y_test, y_pred_03)

## Display the confusion matrix
print("LR Confusion Matrix (Threshold = 0.3)")
print("_____")
print(cm_03)
```

LR Confusion Matrix (Threshold = 0.3)

```
[[2485 4596]
 [ 82  642]]
```

```
In [19]: ## Create and display the classification report with 0.3 threshold
print("LR Classification Report (Threshold = 0.3)")
print("_____")
print(classification_report(y_test, y_pred_03))
```

LR Classification Report (Threshold = 0.3)

	precision	recall	f1-score	support
0	0.97	0.35	0.52	7081
1	0.12	0.89	0.22	724
accuracy			0.40	7805
macro avg	0.55	0.62	0.37	7805
weighted avg	0.89	0.40	0.49	7805

Figure 14

Random Forest AUC and Confusion Matrix Values

```
Out[20]: RandomForestClassifier(class_weight='balanced', min_samples_leaf=5,  
                                min_samples_split=10, n_estimators=200, n_jobs=-1,  
                                random_state=123)
```

**In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.**

```
In [21]: # Make predictions  
y_pred_rf = rf.predict(X_test)  
y_prob_rf = rf.predict_proba(X_test)[:, 1]  
  
## Calculate RF AUC value  
auc_rf = roc_auc_score(y_test, y_prob_rf)  
  
## Display RF AUC value  
print("RF AUC:", auc_rf)
```

RF AUC: 0.7491112119351373

```
In [22]: ## Create a confusion matrix for the RF model  
cm_rf = confusion_matrix(y_test, y_pred_rf)  
  
## Display the results  
print("RF Confusion Matrix (Threshold - 0.5)")  
print("_____")  
print(cm_rf)
```

RF Confusion Matrix (Threshold - 0.5)

```
[[6013 1068]  
 [ 363  361]]
```

Figure 15

Random Forest ROC Curve

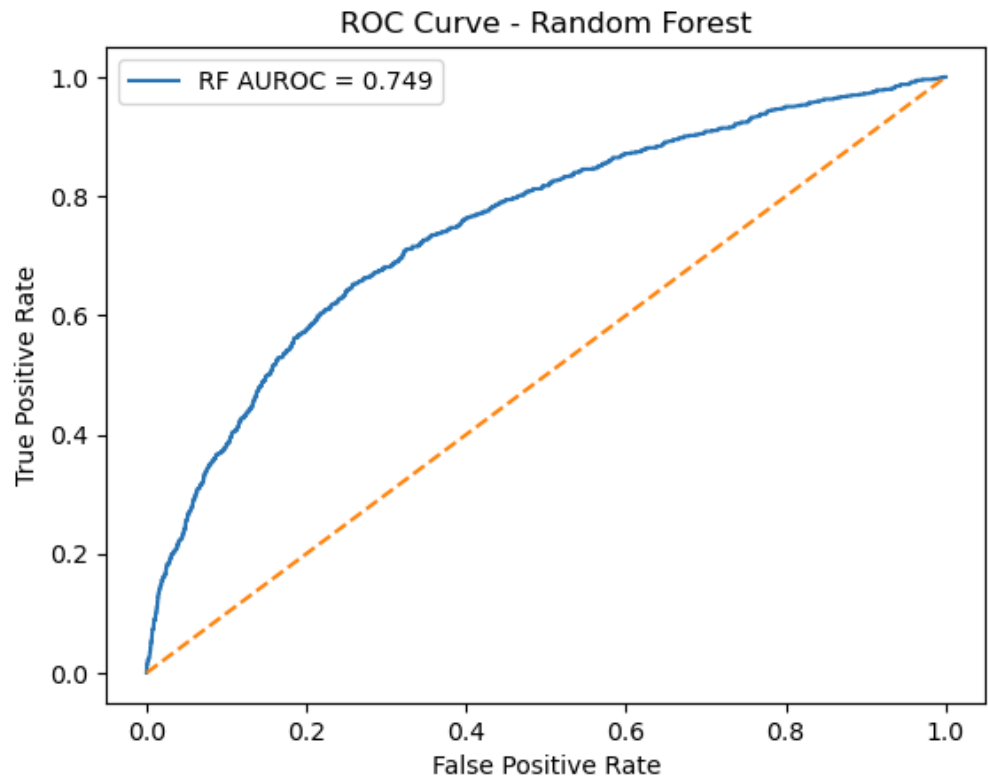


Figure 16

Random Forest Classification Report

```
In [23]: ## Create and display the classification report for the RF model
print("RF Classification Report (Threshold = 0.5)")
print("_____")
print(classification_report(y_test, y_pred_rf))
```

RF Classification Report (Threshold = 0.5)

	precision	recall	f1-score	support
0	0.94	0.85	0.89	7081
1	0.25	0.50	0.34	724
accuracy			0.82	7805
macro avg	0.60	0.67	0.61	7805
weighted avg	0.88	0.82	0.84	7805

Figure 17

Random Forest Feature Importance Table

	RF Feature	Importance
0	age	0.208317
22	ip_admit_utilization_moderate	0.129341
5	los_days_capped	0.118202
17	med_risk_high	0.088668
21	ip_admit_utilization_low	0.060224
23	ip_admit_utilization_high	0.044922
16	med_risk_medium	0.034747
3	dispo_transfer	0.033566
1	gender_female	0.029165
18	ed_low	0.026927
19	ed_moderate	0.026815
20	ed_high	0.024473
14	mort_risk_high	0.021530
13	mort_risk_medium	0.019755
24	icu_admit_flag	0.019520
2	dispo_post_acute	0.018051
15	mort_risk_missing	0.016488
6	infection_flag	0.014405
25	rrt_flag	0.014216
11	adm_dx_rf	0.009538
12	adm_dx_aki	0.008914
8	adm_dx_pna	0.008608
4	dispo_ama	0.007939
9	adm_dx_hf	0.006062
7	adm_dx_sepsis	0.005371
10	adm_dx_copd	0.004237

Figure 18

Gradient Boosting (XGBoost) AUC and Confusion Matrix Values

```
Out[26]: XGBClassifier(base_score=None, booster=None, callbacks=None,
                      colsample_bylevel=None, colsample_bynode=None,
                      colsample_bytree=0.8, device=None, early_stopping_rounds=None,
                      enable_categorical=False, eval_metric='logloss',
                      feature_types=None, feature_weights=None, gamma=None,
                      grow_policy=None, importance_type=None,
                      interaction_constraints=None, learning_rate=0.05, max_bin=None,
                      max_cat_threshold=None, max_cat_to_onehot=None,
                      max_delta_step=None, max_depth=3, max_leaves=None,
                      min_child_weight=None, missing=nan, monotone_constraints=None,
                      multi_strategy=None, n_estimators=300, n_jobs=None,
                      num_parallel_tree=None, ...)
```

**In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.**

```
In [27]: ## Make predictions
y_pred_gb = gb.predict(X_test)
y_prob_gb = gb.predict_proba(X_test)[:, 1]

## Calculate the AUC value
auc_gb = roc_auc_score(y_test, y_prob_gb)

## Display the AUC value
print("GB AUC:", auc_gb)
```

GB AUC: 0.7559203447713553

```
In [28]: ## Create the GB confusion matrix
gb_cm = confusion_matrix(y_test, y_pred_gb)

## Display the GB confusion matrix
print("GB Confusion Matrix (Threshold = 0.5)")
print("_____")
print(gb_cm)
```

GB Confusion Matrix (Threshold = 0.5)

```
[[5631 1450]
 [ 295  429]]
```

Figure 19

Gradient Boosting Classification Report

```
In [29]: ## Create and display the GB classification report
print("GB Classification Report (Threshold = 0.5)")
print("_____")
print(classification_report(y_test, y_pred_gb))
```

GB Classification Report (Threshold = 0.5)

	precision	recall	f1-score	support
0	0.95	0.80	0.87	7081
1	0.23	0.59	0.33	724
accuracy			0.78	7805
macro avg	0.59	0.69	0.60	7805
weighted avg	0.88	0.78	0.82	7805

Figure 20

Random Forest and Gradient Boosting (XGBoost) ROC Curve Plots

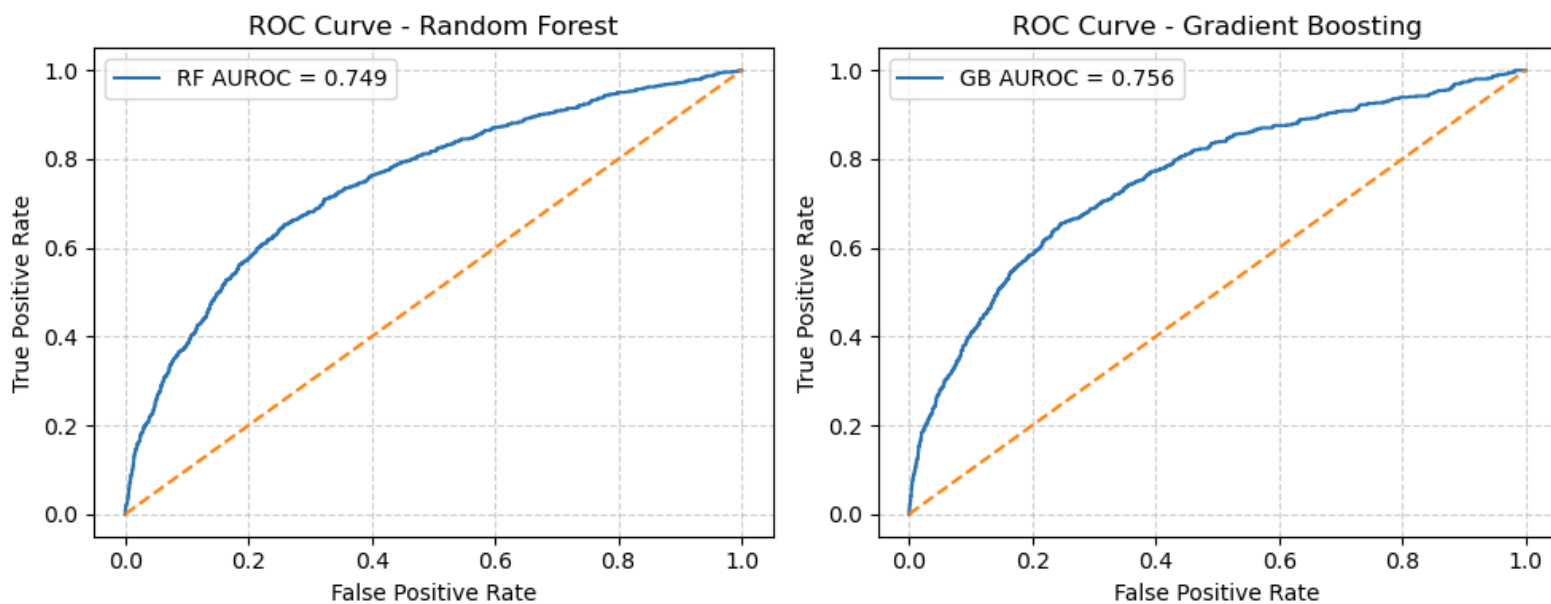


Figure 21

Gradient Boosting Feature Importance (Gain) Chart

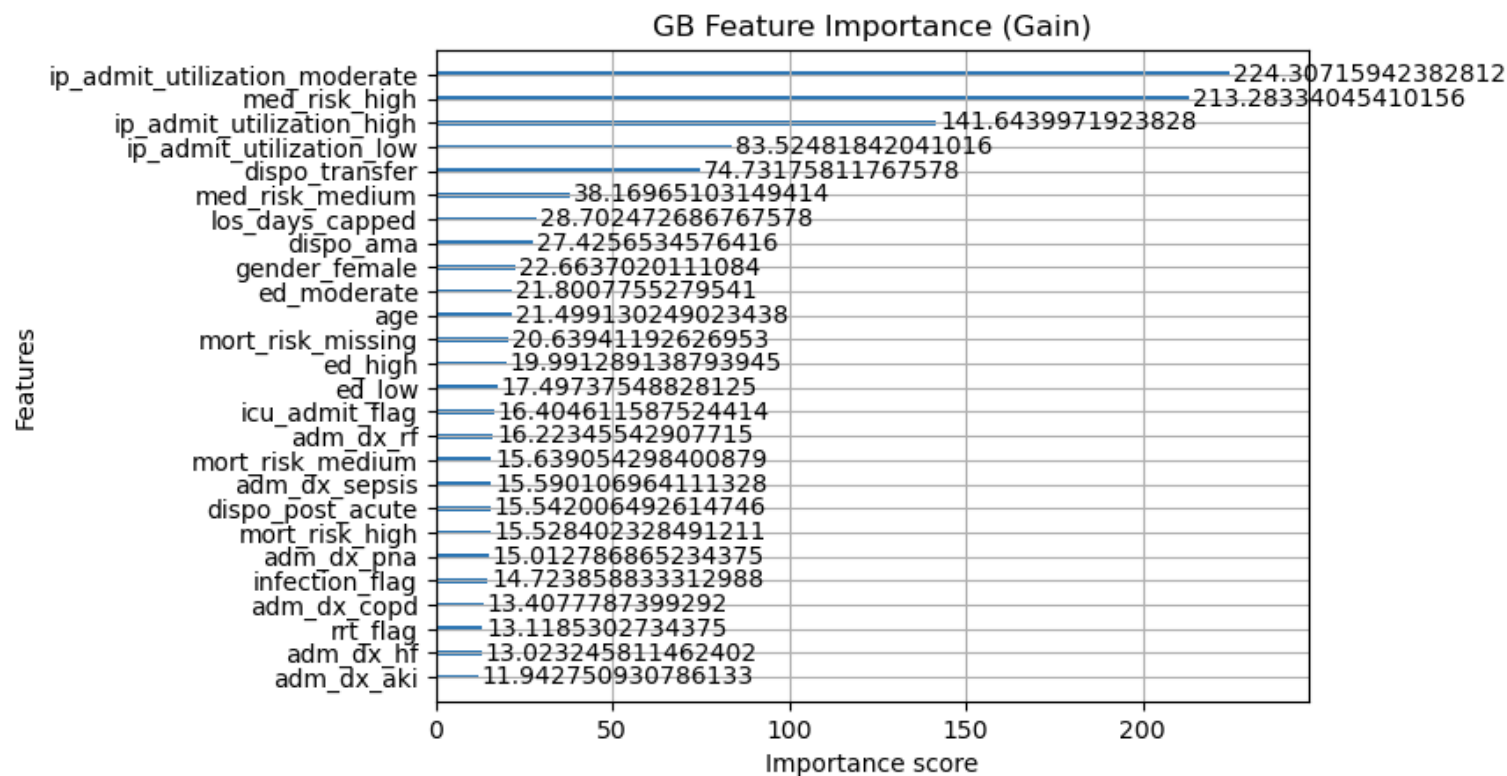


Figure 22

DeLong's Test for Correlated ROC Curves Results

```
In [33]: ## Run DeLong's Test: LR vs RF
dlt_lr_rf = delong_roc_test(y_test, y_prob, y_prob_rf)

## Run DeLong's Test: LR vs GB
dlt_lr_gb = delong_roc_test(y_test, y_prob, y_prob_gb)

## Display DeLong's Test results
print("DeLong's Test - AUROC Comparisons")
print("_____")
print("LR vs RF p-value:", dlt_lr_rf)
print("LR vs GB p-value:", dlt_lr_gb)
```

DeLong's Test - AUROC Comparisons

LR vs RF p-value: 0.6991787880344789
LR vs GB p-value: 0.06644610838854659